

ITM SLS BARODA UNIVERSITY, VADODARA
B.TECH. SEMESTER-3 (BRANCH – ALL; SCHOOL – SOCSET)
DEPARTMENT OF APPLIED MATHEMATICS (S.O.S.)
COMPUTER ORIENTED NUMERICAL METHODS WITH PYTHON (S2320C1)

CET-2 SYLLABUS

Topic – Interpolation Techniques

- ▶ Definition: Interpolation and Extrapolation
- ▶ Operators:
 - Types of Operators
 - Relation between operators
- ▶ Finite Differences along with tables:
 - Forward, Backward and Central Differences
- ▶ Methods for Equally Spaced Intervals:
 - Newton's Forward Difference Interpolation Method
 - Newton's Backward Difference Interpolation Method
 - Stirling's Method
- ▶ Methods for Unequally Spaced Intervals:
 - ▶ Lagrange's Interpolation Method

Topic – Errors and Approximations

- ▶ Errors
 - Absolute, Relative, Percentage Errors

Topic – Solution to System of Linear Equation

- ▶ Iterative Methods
 - Jacobi Method
 - Seidel Method